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(54) **DATA BUS WIDTH CONFIGURABLE INTERCONNECTION CIRCUITRY**

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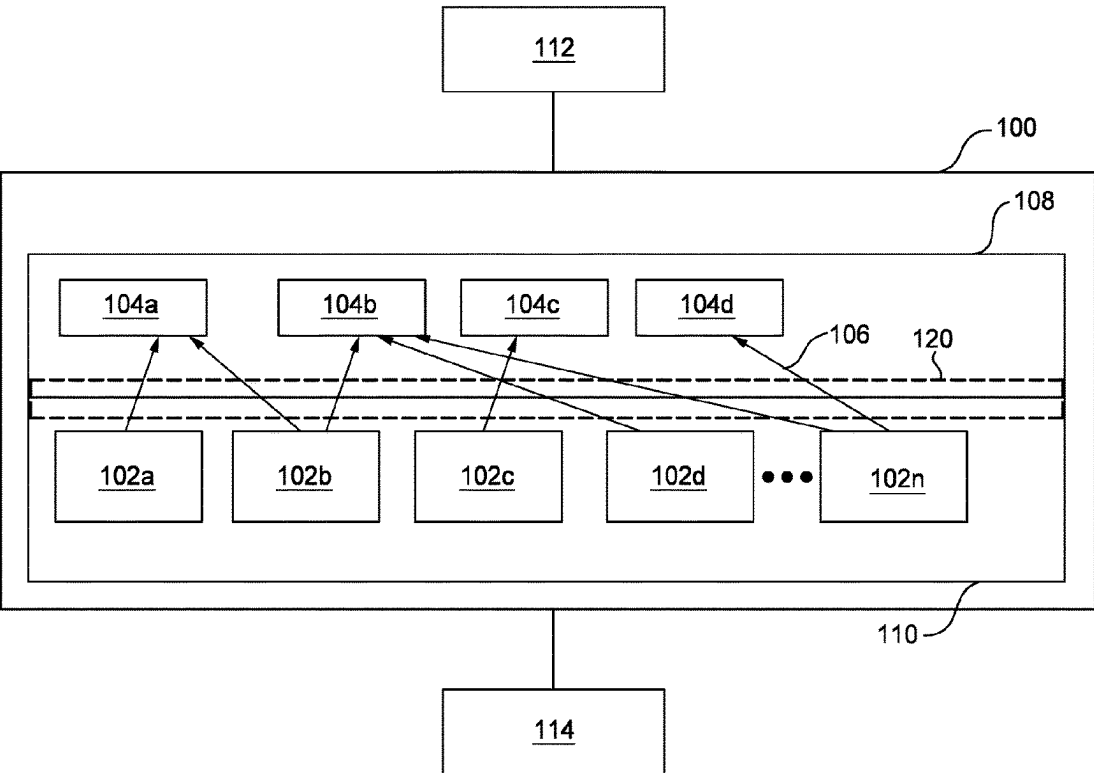
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(57) **ABSTRACT**
An interconnection circuitry of an accelerator device includes a multiplexer, a first plurality of buffers, a second plurality of buffers, and a demultiplexer. The multiplexer is coupled to first offload circuitry and received data therefrom. The first plurality of buffers has inputs coupled to outputs of the multiplexer. A second plurality of buffers has inputs coupled to outputs of the first plurality of buffers. The demultiplexer includes inputs coupled to outputs of the second plurality of buffers and outputs coupled to inputs of programmable logic.

20 Claims, 7 Drawing Sheets



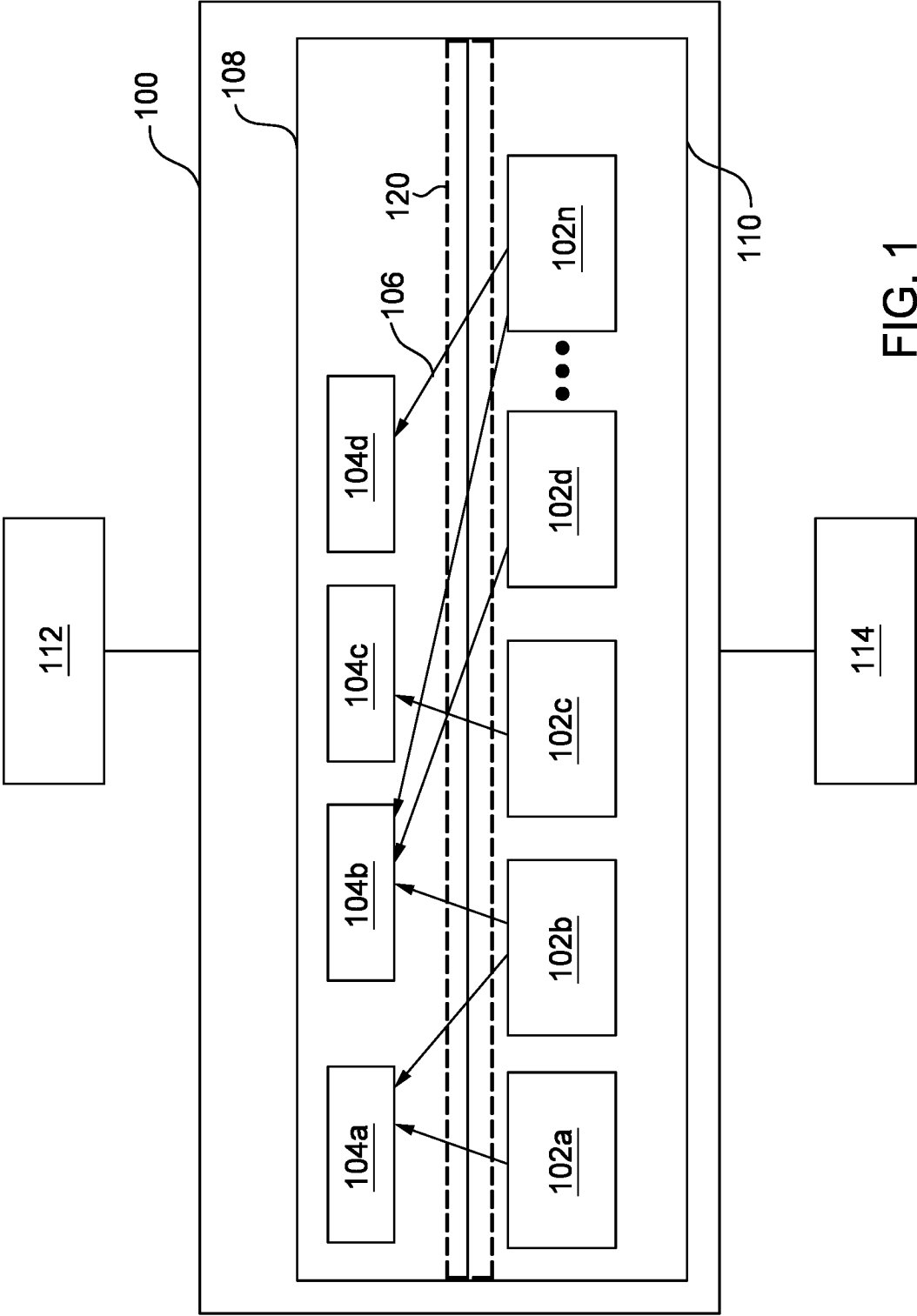


FIG. 1

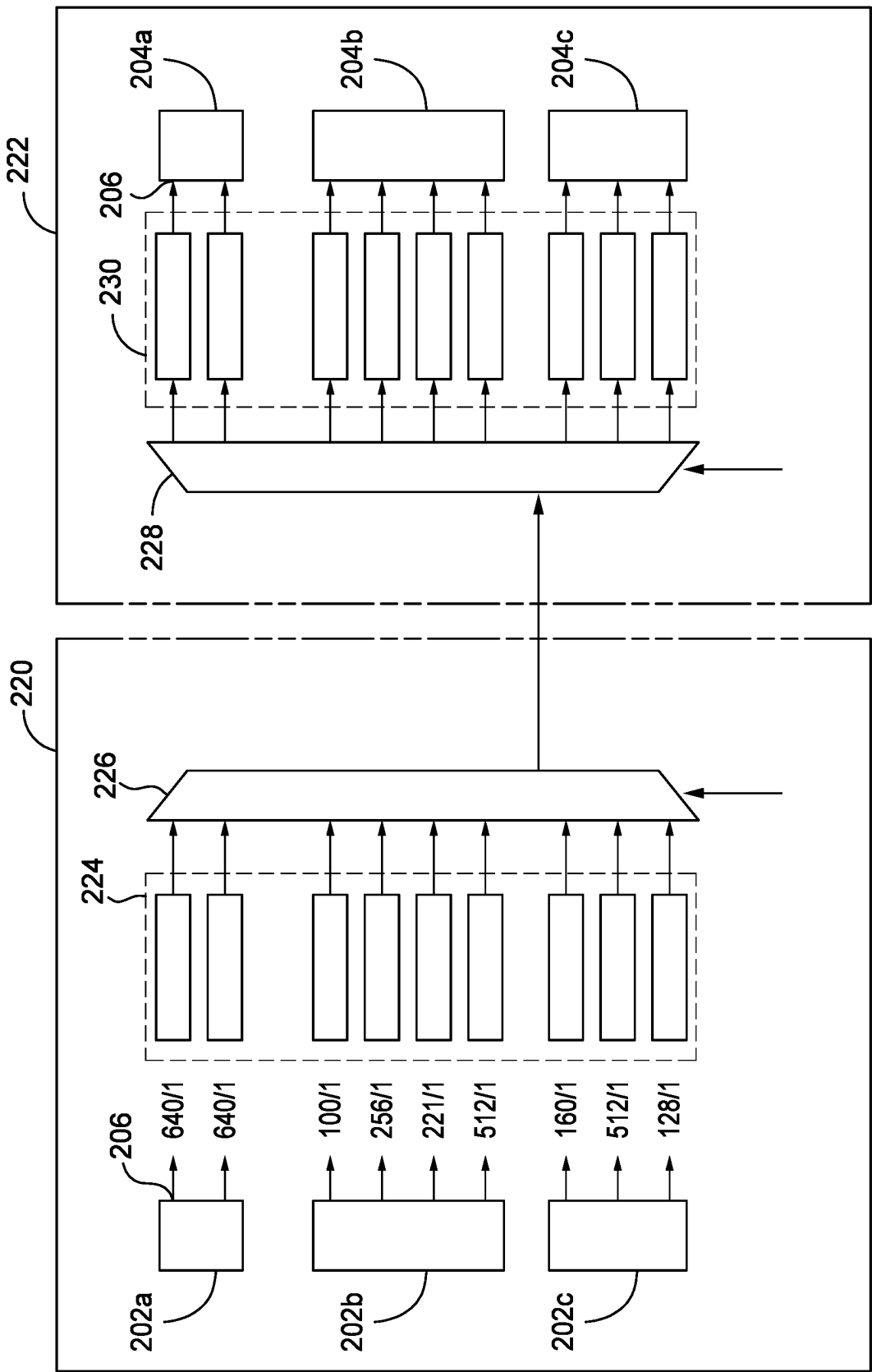


FIG. 2 (PRIOR ART)

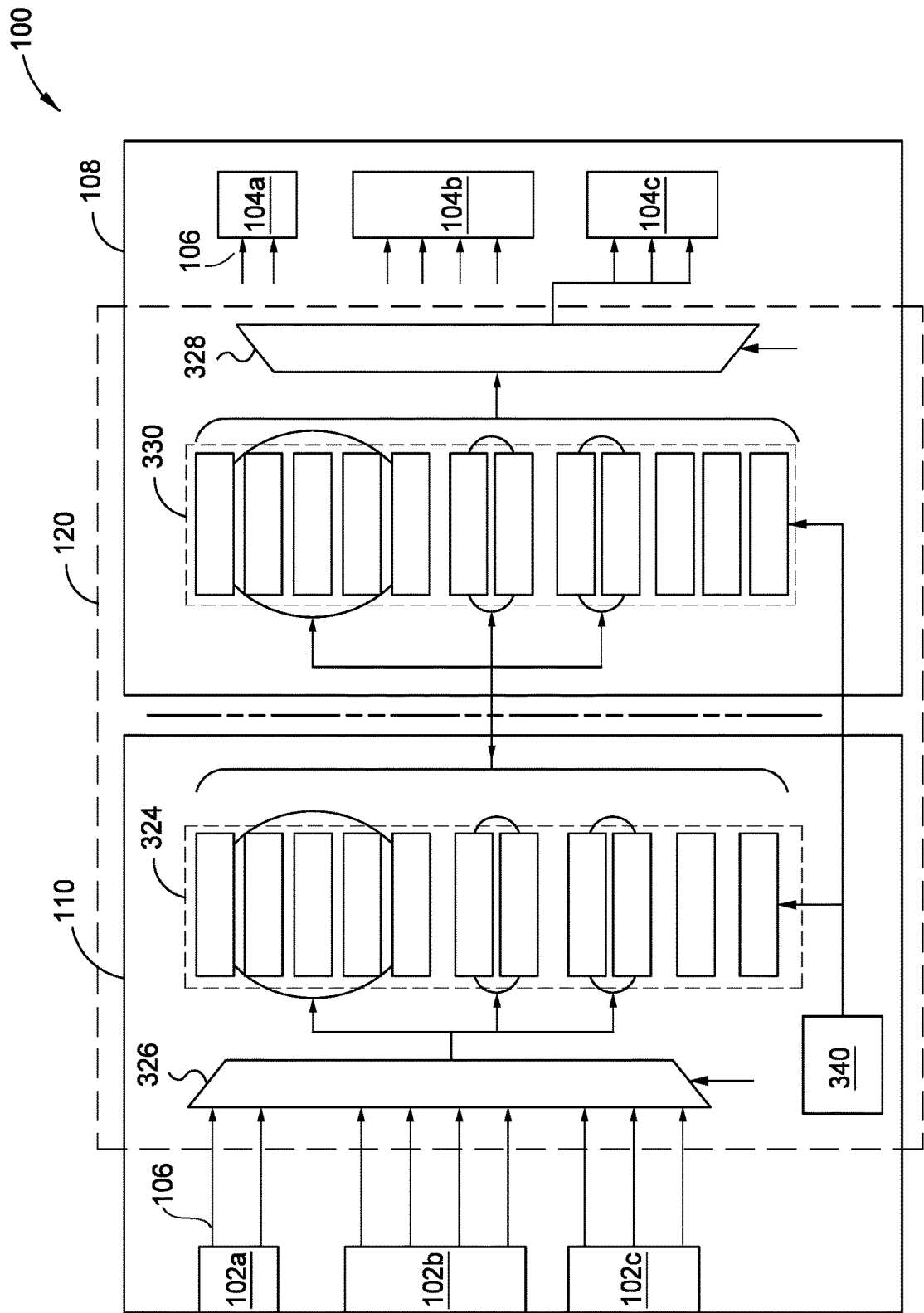


FIG. 3

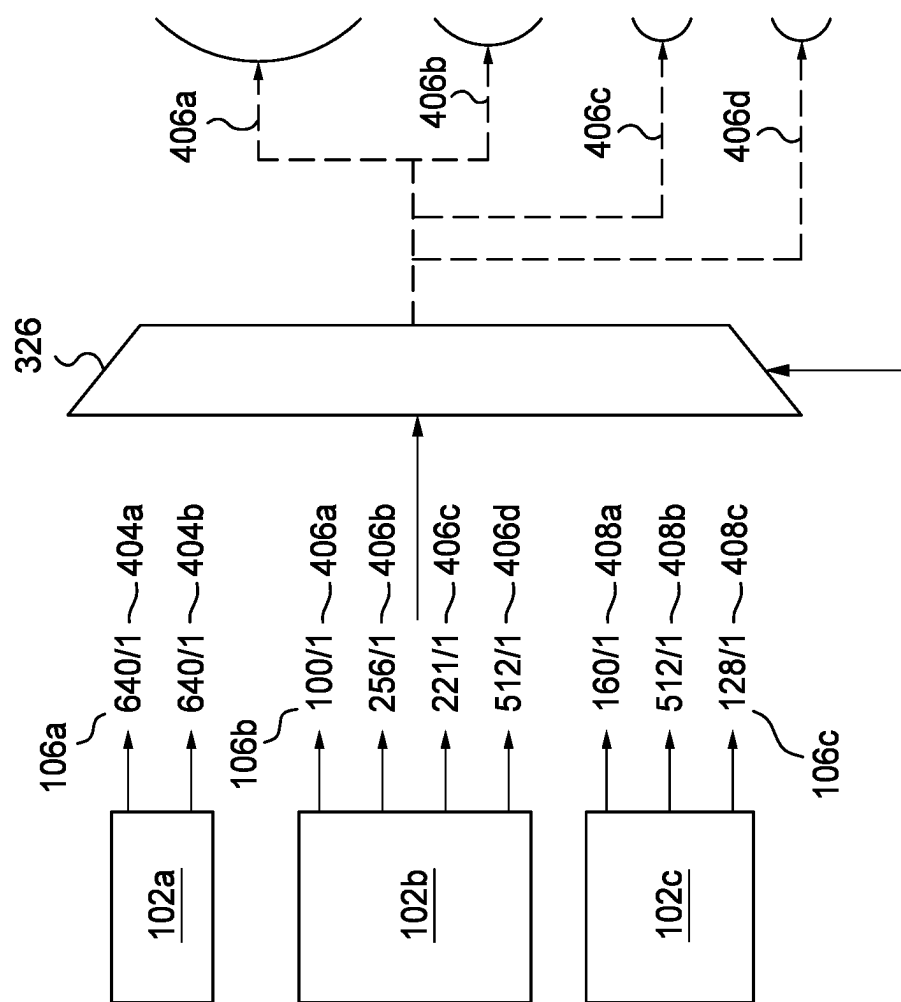


FIG. 4

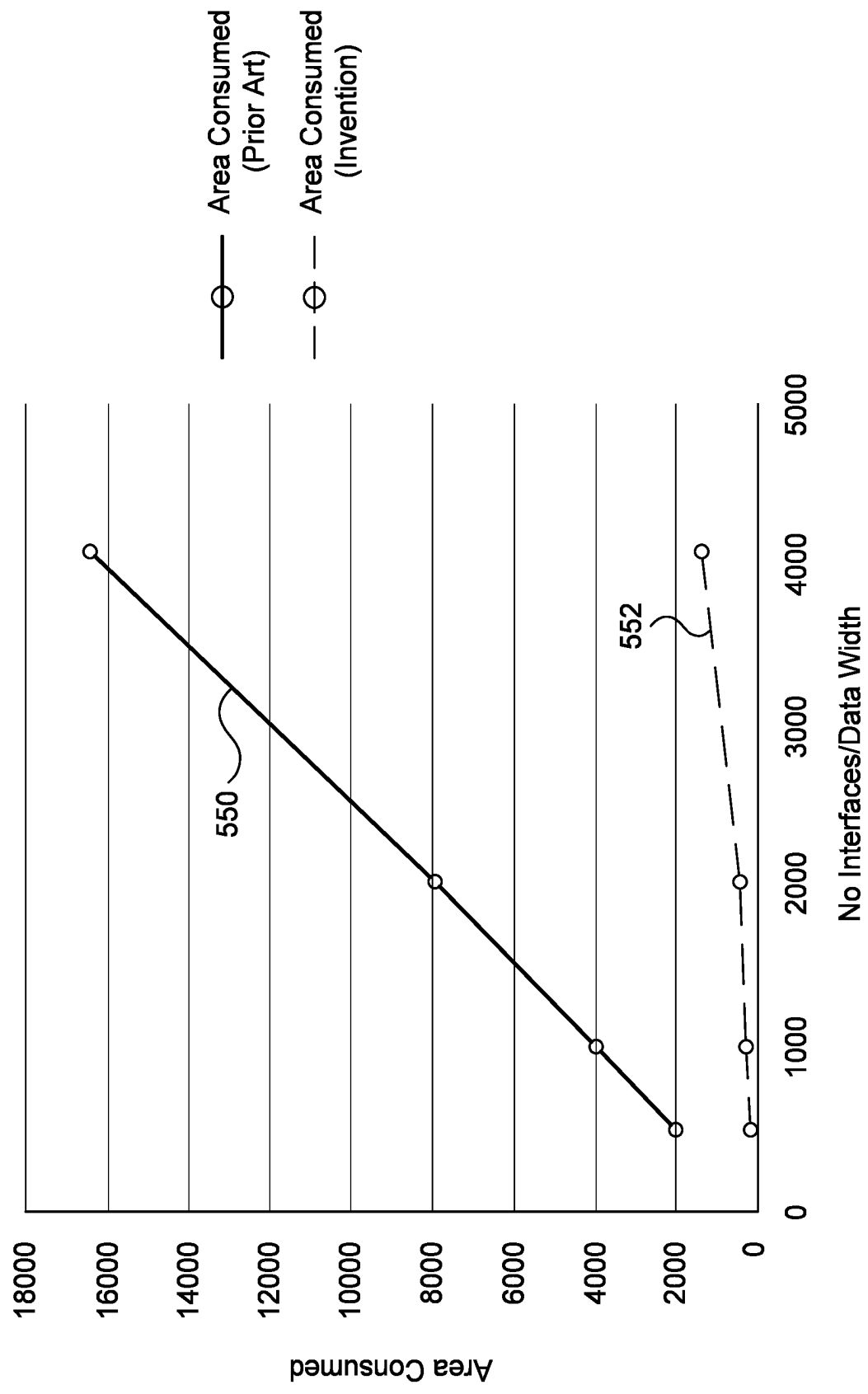


FIG. 5

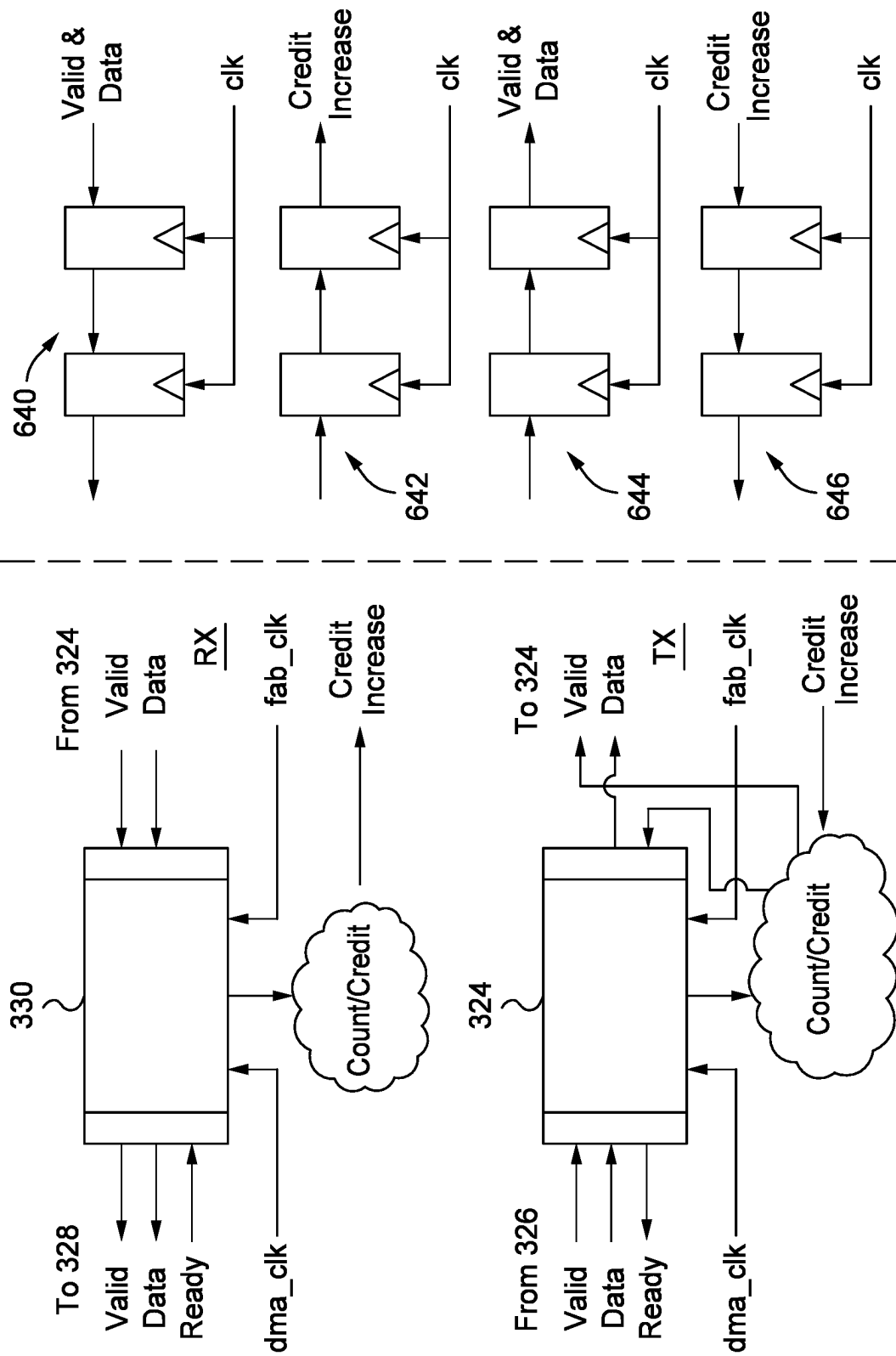


FIG. 6

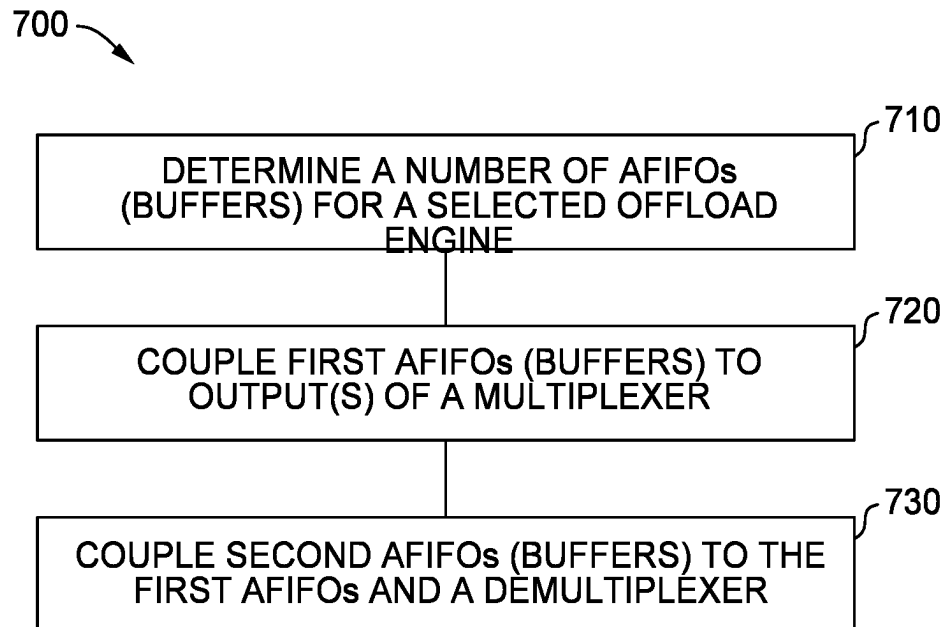


FIG. 7

1

DATA BUS WIDTH CONFIGURABLE INTERCONNECTION CIRCUITRY

TECHNICAL FIELD

Embodiments of the present disclosure generally relate to accelerator devices for offloading processing tasks, and in particular, to flexible interconnection circuitry within an accelerator device that is able to support various bus data widths.

BACKGROUND

Intelligent server adapters, or smart network interface controllers (SmartNICs) or accelerator devices, boost server performance in cloud and data centers by offloading central processing unit (CPU) tasks, such as network and/or data processing. Intelligent server adapters accelerate network and/or storage intensive tasks by executing them directly on the intelligent server adapters independently of the software-controlled server CPUs, thereby eliminating the need to run these processing workloads on servers and freeing up CPU cycles, thus improving server performance and reducing overall power consumption, resulting in lower system-level total cost of ownership. Intelligent server adapters improve effective network bandwidth and throughput by executing offloaded functions on the faster processing hardware instead of the slower software driven server CPUs.

SUMMARY

In one example, an interconnection circuitry includes a multiplexer, a first plurality of buffers, a second plurality of buffers, and a demultiplexer. The multiplexer is coupled to first offload circuitry and received data therefrom. The first plurality of buffers has inputs coupled to outputs of the multiplexer. A second plurality of buffers has inputs coupled to outputs of the first plurality of buffers. The demultiplexer includes inputs coupled to outputs of the second plurality of buffers and outputs coupled to inputs of programmable logic.

In one example, an accelerator device includes first offload circuitry, and interconnection circuitry. The interconnection circuitry is coupled to the first offload circuitry. The interconnection circuitry includes a multiplexer, a first plurality of buffers, a second plurality of buffers, and a demultiplexer. The multiplexer is coupled to the first offload circuitry. The first plurality of buffers has inputs coupled to outputs of the multiplexer. The second plurality of buffers has inputs coupled to outputs of the first plurality of buffers. The demultiplexer has inputs coupled to outputs of the second plurality of buffers and outputs coupled to inputs of programmable logic.

In one example, a method includes determining, based on a data bus width of a first offload circuitry, a number of a first plurality of buffers and a number of a second plurality of buffers. Further, the method includes coupling inputs of the first plurality of buffers to outputs of a multiplexer. Inputs of the multiplexer are coupled to outputs of the first offload circuitry. The method further includes coupling inputs of the second plurality of buffers to outputs of the first plurality of buffers and coupling outputs of the second plurality of buffers to inputs of demultiplexer. Outputs of the data demultiplexer are coupled to inputs of programmable logic.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present invention can be understood in detail, a more

2

particular description of the invention, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.

FIG. 1 illustrates a schematic block diagram of an accelerator device.

FIG. 2 illustrates a schematic block diagram of prior art fabric multiplexer and demultiplexer circuits.

FIG. 3 illustrates a schematic block diagram of interconnection circuitry.

FIG. 4 illustrates a schematic block diagram of a configuration of AFIFOs for an accelerator device.

FIG. 5 illustrates a schematic comparison graph of circuit area of different interconnection circuitries.

FIG. 6 illustrates a schematic logic block diagram of credit-based data transfer circuits.

FIG. 7 illustrates a flowchart of a method for coupling AFIFOs to a multiplexer and demultiplexer.

To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures, and a lower-case letter added where the elements are substantially the same. It is contemplated that elements of one embodiment may be beneficially incorporated in other embodiments.

DETAILED DESCRIPTION

Data centers (e.g., cloud computing centers and other distributed data processing centers) employ multiple computer processing systems to perform various distributed computing processes. The computing processing systems include one or more central processing units (CPUs). In many instances, data centers further include accelerator devices (e.g., intelligent server adapters, or smart network interface controllers (SmartNICs)). The accelerator devices offload CPU tasks, improving the performance of the data centers. In one example, the tasks include network processing tasks and/or data processing tasks. The accelerator devices reduce processing performed by a software driven server CPU by offloading processor intensive tasks from the CPU. Offloading the processor intensive tasks frees up CPU cycles, improving server performance and reducing overall power consumption, resulting in lower system-level total cost of ownership of a data center.

An accelerator device improves the effective network bandwidth and throughput by executing offloaded functions within the faster processing hardware of the accelerator device instead of within slower software driven server CPUs. An accelerator device may be implemented as a peripheral device that is coupled to a host device (e.g., a computing device) to offload tasks from the CPU. In one or more examples, multiple accelerator devices may be coupled to a host device. Further, an accelerator device may offload tasks from one or more CPUs of a host device. In one example, multiple accelerator devices may offload tasks from one or more CPUs of a host device.

An accelerator device includes one or more offload circuitries and programmable logic. The offload circuitries are coupled with the programmable logic to perform tasks offloaded from a corresponding CPU. In an accelerator device interface connections are used to connect offload circuitries with the programmed portions of the programmable logic. In one example, the programmable logic includes programmed portions associated with different user

applications, which are connected to the offload circuitries via the interface connections of interconnection circuitry. However due to the limited availability of the signal connections (pins) within the interconnection circuitry, a conventional accelerator device shares signal pins, and, consequently, multiplexes the functionality of the accelerator device on the pins.

The accelerator device described in the following employs asynchronous first-in-first-out shift registers (AFIFOs) to connect the offload circuitries with programmed circuit blocks of the programmable logic. The AFIFOs are grouped according to the data bus width of the offload circuitries. For example, the AFIFOs may be grouped into multiple groups, where each group has a data bus width at least as large as the data bus width of a respective offload circuitry. Further, as different offload circuitries are selected to be connected to different programmed circuit blocks, the AFIFOs are regrouped to accommodate the different data bus widths. Grouping together multiple AFIFOs to support the data bus widths of the offload circuitries, allows for the circuit size of the AFIFOs to be reduced as compared to conventional interconnection circuitry. Further, grouping of AFIFOs as described herein provides support for various data bus widths, accommodating offload circuitries with different data bus widths and reducing unused data bus width within the interconnection circuitry. Accordingly, the circuit size and manufacturing cost of the corresponding semiconductor device is reduced.

FIG. 1 illustrates a block diagram of an accelerator device **100**, according to one or more examples. In one example, the accelerator device **100** is a field-programmable gate array (FPGA) device. For example, an accelerator device may be an embedded FPGA (eFPGA) peripheral device.

The accelerator device **100** is coupled to one or more host devices **114**. For example, the accelerator device **100** is coupled to the host device **114** via an interconnect interface (e.g., a Peripheral Component Interconnect Express (PCIe) or another type interconnect interface). The accelerator device **100** is further coupled to one or more remote hosts **112**. The accelerator device **100** may be coupled to the remote hosts **112** via a network connection. The accelerator device **100** may include a network interface that is able to communicate directly with the remote hosts **112** via the network connection. In another example, the accelerator device **100** communicates with the remote hosts **112** via network interface of the host device **114**.

The host device **114** includes one or more CPUs, one or more memory devices, and interface devices. The host device **114** may be coupled to the remote hosts **112** via a network interface. The host device **114** performs one or more processing functions and provides processed data to the remote hosts **112**. The remote hosts **112** receive processed data from one or more host devices **114**. The remote hosts **112** combine the process data received from two or more host devices **114** and/or further process the data received from the host devices **114**. Further, the remote hosts **112** may communicate data received from one or more host devices **114** to another one or more host devices **114**. The host device **114** and the remote hosts **112** are part of a distributed computing system. In one example, the remote hosts **112** are configured similar to the host device **114**.

The offload circuitries **102** are formed within processor circuitry **110** (e.g., hardened circuitry) of the accelerator device **100**. In one example, the processor circuitry **110** is application specific integrated circuit (ASIC) processor circuitry, and an offload circuitry **102** is an ASIC offload circuitry. In one or more examples, the offload circuitries

102 may be, for example but not limited to, direct memory access (DMA) engines, Data Splitters (like based on MPS), Data Encryption/Decryption engines, Data Compression/Decompression engines, and the like. The offload circuitries **102** perform tasks offloaded from a CPU within the host device **114**, freeing up the CPU to perform other tasks. For example, the host device **114** uses the offload circuitries **102** to perform data splitting, data compression, and/or data encryption.

The programmed circuit blocks **104** are part of the programmable logic **108** of the accelerator device **100**. The programmed circuit blocks **104** are generated within the programmable logic **108** based on configuration data. The configuration data may correspond to an application (e.g., a user application). Each of the programmed circuit blocks **104** may be programmed based on a different application (or portion of an application), and corresponding configuration data. Accordingly, two or more of the programmed circuit blocks **104** are configured to perform a different task or a different portion of a common task.

In one example, the programmable logic **108** includes circuitry that can be configured to implement various processing tasks. Functionality of the programmable logic **108** is programmed via configuration data. In one example, altering the functionality of the programmable logic **108** alters the interconnections between logic elements within the programmable logic **108** based on the configuration data, forming the programmed circuit blocks **104**. The processor circuitry **110** is hardened logic circuitry that has a predetermined functionality (e.g., filters, mixers, sample-rate converters, and transform circuits, among others). The processor circuitry **110** may be programmable to configure specific predetermined functionalities or select among predetermined functionalities. However, in contrast to the programmable logic **108**, the processor circuitry **110** cannot be configured or reconfigured with different functionality (e.g., additional functionalities cannot be added to or removed from the processor circuitry **110**). In contrast, programmable logic can be reconfigured to add one more additional functionalities or to remove a functionality.

The offload circuitries **102** are coupled to the programmed circuit blocks **104** via the interconnection circuitry **120**. The interconnection circuitry **120** includes one or more multiplexers, one or more demultiplexers, and AFIFOs. The multiplexers, the demultiplexers, and the AFIFOs form the interfaces **106** that connect the offload circuitry **102** with the programmed circuit blocks **104**. As will be described in greater detail in the following, the AFIFOs are coupled between the multiplexers and demultiplexers, and are grouped based on the data bus width of each of the offload circuitries **102**, to provide the interfaces **106**.

Referring to FIG. 2, depicted is a block diagram of prior art fabric multiplexer and demultiplexer circuits. The hardened circuitry **220** and programmable logic **222** each use dedicated AFIFOs to provide the bandwidth, latency, timing, and clock domain crossing for each interface to send and/or receive data between offload circuitries **202a**, **202b** and **202c** within the hardened circuitry **220** and the programmed circuit blocks **204a**, **204b**, and **204c** of the programmable logic **222** by using valid/ready handshake signals to control the data flow. Each interface **206** may have one or more ready/valid handshake interfaces with data widths {640, 640}, {100,256,221,512}, {160,512,128} bits wide. Therefore, dedicated AFIFOs **224**, **230**, having the same data bus widths as each interface **206** are provided. The respective AFIFOs **224**, **230**, and the interfaces **206** are selected with a multiplexer **226** and demultiplexer **228**. The AFIFOs **224**,

are required to be configured to handle full data bus widths for each interface **206**. Accordingly, there is a need for a large number of AFIFOs that meet the full data bus width requirements of each associated offload circuitry **202**. Having dedicated AFIFOs for each interface **206** increases the amount of circuit area used by a corresponding integrated circuit (IC) device, and the circuit area used by the AFIFOs **224**, **230** increases as the number of interfaces **206** and/or the corresponding number of interfaces and/or corresponding data bus widths increase.

Thus, there is a need for reducing the number of AFIFOs and corresponding circuit area used for data transfer to and from high-speed data processing peripherals, e.g., offload circuitries **102**, while achieving expected bandwidth, latency, timing, and clock domain crossings.

In the example of FIG. 3, instead of requiring a large data width AFIFO for each interface **106**, a plurality of AFIFOs **324**, **330** having a smaller data width than that used in the example of FIG. 2, e.g., data width equal to 16/32/48 bits wide, are grouped together to provide the necessary data width of the interfaces **106**. The AFIFOs **324**, **330** are grouped to increase the combined data width of the grouped AFIFOs to at least match the corresponding interface **106**. These AFIFO **324**, **330** are connected to the output of the multiplexer of the processor circuitry **110** (e.g., the multiplexer **326** of FIG. 3), and to the input of the demultiplexer of the programmable logic **108** (e.g., the demultiplexer **328** of FIG. 3), improving the bandwidth, latency, timing, and clock domain crossings of the interfaces **106**.

Thus, the total number of AFIFOs within an accelerator device (e.g., the accelerator device **100** of FIG. 1) are reduced, reducing the cost of the accelerator device. Further, scaling up to accommodate more offload circuitries (e.g., offload circuitries **102** of FIG. 1) does not significantly impact the circuit area used by the AFIFOs (e.g., AFIFOs **324**, **330**), because existing AFIFO are reused and regrouped for each offload circuitry interface selected for data transfer. Referring to FIG. 5, depicted is a schematic comparison graph of circuit area used by the prior art solution of FIG. 2, represented by graph line **550**, and the circuit area when using repurposed (reconfigured) AFIFOs at the outputs of the multiplexer and demultiplexers as described with regard to FIG. 3, represented by graph line **552**. As is illustrated in FIG. 5, as the data width increases, the circuit area when using repurposed AFIFOs as described herein is less than that of the circuit area of a prior art solution of FIG. 2. Further, as is described herein, using repurposed AFIFOs allows for the same performance and latency to be achieved while using significantly less circuit area and number of AFIFO circuits, as compared to a convention accelerator device.

Examples of the present disclosure relate to efficient and cost-effective transfer of data between accelerator devices (e.g., accelerator device **100** of FIG. 1), host devices (e.g., host device **114** of FIG. 1) and remote hosts (e.g., remote hosts **112** of FIG. 1). As is described above, accelerator devices offer offload functionalities and provide different interfaces that are able to serve the unique needs of a computer data processing system. However due to the limited availability of signal connections (pins) within the interconnection circuitry of a conventional accelerator device, the pins are shared and multiplexing is used communicate between the offload circuitries and the programmed circuit blocks. As is described in greater detail in the following, a wide data width multiplexer, a wide data width demultiplexer, and a plurality of narrow data width AFIFOs are used to provide interconnection between offload

circuitries and programed circuit blocks. A wide data width multiplexer and demultiplexer have a data width of 256 bits, 512 bits, 1024 bits, or greater. In other examples, a multiplexer and/or demultiplexer having a data width of less than 256 bits may be used. The AFIFOs are grouped together and coupled to the wide data width multiplexer and demultiplexer to provide data transfers over a wide data width bus, mitigating the sharing of pins and multiplexing within the accelerator device.

According to examples disclosed herein, a multiplexer selects an interconnection based upon operational requirements of the corresponding offload circuitry and programmed circuit blocks, providing clock domain crossing functionality between a processor circuitry clock domain to a programmable logic clock domain. In one example, data is provided to (e.g., pushed) to the AFIFOs based on a first clock signal (e.g., in a first clock domain) and output from (e.g., popped) from the AFIFOs based on a second clock signal (e.g., in a second clock domain). For example, data is provided from the offload circuitry **102a** via the multiplexer **326** to a first group of the AFIFOs **324** during a cycle of a clock signal associated with the processor circuitry **110**. Further, the data is output from the AFIFOs **324** in a cycle of a clock signal associated with the programmable logic **108**. Accordingly, the data crosses over from the clock domain of the processor circuitry **110** to the clock domain of the programmable logic **108**. In one or more examples, the frequency of the clock domain of the processor circuitry **110** is greater than or less than the frequency of the clock domain of the programmable logic **108**. The data is loaded into the AFIFOs **324** in one clock cycle of a first clock signal, and the data is output from the AFIFOs **324** in one clock cycle of a second clock signal. In one or more examples, the interconnection circuitry (e.g., the interconnection circuitry **120** of FIG. 1) operates at a frequency of at least 433 MHZ, and uses at least sixteen AFIFOs. In other examples, other operating frequencies and/or more than or less than sixteen AFIFOs may be used. In one or more examples, credit-based mechanisms are used to meet bandwidth and timing requirements. Further, head of line blocking (HOL) blocking corresponding to a multiplexer is avoided as all buffering is handled at the outputs of the multiplexer.

Referring to FIG. 3, depicted is a block diagram of the interconnection circuitry **120** of the accelerator device **100** of FIG. 1, according to one or more examples. The interconnection circuitry **120** includes multiplexer **326**, AFIFOs **324**, AFIFOs **330**, and demultiplexer **328**.

The interconnection circuitry **120** is partially formed within the processor circuitry **110** and the programmable logic **108**. The interconnection circuitry **120** couples the offload circuitries **102** with the programmed circuit blocks **104**. In one example, the multiplexer **326** and the AFIFOs **324** are formed within the processor circuitry **110**, and the AFIFOs **330** and the demultiplexer **328** are formed within the programmable logic **108**. The data width of the multiplexer **326** is greater than that of each of the AFIFOs **324**. Further, the data width of the demultiplexer **328** is greater than that of each of the AFIFOs **330**.

In one example, data is output from the offload circuitries **102** and received by the programmed circuit blocks **104** via the multiplexer **326**, the AFIFOs **324**, the AFIFOs **330**, and the demultiplexer **328**. In one example, inputs of the multiplexer **326** are coupled to outputs of the offload circuitries **102**. Output or outputs of the multiplexer **326** are coupled to inputs of the AFIFOs **324**. Outputs of the AFIFOs **324** are coupled to input to inputs of the AFIFOs **330**. Outputs of the AFIFOs **330** are connected to inputs of the demultiplexer

328. Outputs of the demultiplexer **328** are coupled to inputs the programmed circuit blocks **104**.

The AFIFOs **324**, **330** have a data width of about 16/32/48 bits wide. By locating the AFIFOs **324** at the outputs of the multiplexer **326**, and locating the AFIFOs **330** at the inputs of the demultiplexer **328**, the opposite of what is shown in the prior art solution of FIG. 2, the number of AFIFOs **324**, **330** needed to provide for the worst case (e.g., maximum) data width is reduced. In addition to selecting the interface **106** of an offload circuitry **102** and programmed circuit block to be used, the multiplexer **326** and demultiplexer **328** may also select which ones, or groups, of the AFIFOs **324** and **330** are used for the data transfers from/to the selected offload circuitry **102**. In one example, the multiplexer **326** receives a control signal or signals that indicates which offload circuitry **102** to couple with which ones, or groups, of the AFIFO **324**. Further, the demultiplexer **328** receives a control signal or signals that indicates which ones, or groups, of AFIFOs **330** to connect with which programmed circuit block **104**.

In one example, when the offload circuitry **102a** and programmed circuit block **104a** are selected for use, the multiplexer **326** and demultiplexer **328** couples the corresponding interface **106** to a corresponding number of AFIFOs **324**, **330** having a combined data width sufficient to pass the data (at full data width) from the offload circuitry **102a** to the programmed circuit block **104a**. The AFIFOs **324**, **330** are reused (repurposed) and may be selected in any order and number so long as the combined data bus widths of the selected AFIFOs **324**, **330** are sufficient to match the data widths of the selected offload circuitry(s) **102** and corresponding interface(s) **106**. For example, an AFIFO, or group of AFIFOs, **324**, **330** used to provide an interconnection between the offload circuitry **102b** and the programmed circuit block **104b** may additionally be used to provide an interconnection between the offload circuitry **102c** and the programmed circuit block **104c**.

In one example, the accelerator device **100** includes control circuitry **340** that is coupled to the AFIFOs **324** and **330**. The control circuitry **340** groups the AFIFOs **324** and **330** based on the selected offload circuitry (or engine) **102** and the corresponding interface (or interfaces) **106** such that the data bus width of the grouped AFIFOs **324** and **330** is at least at large as that of the interface(s) **106** of the selected offload circuitry(s) **102**. The control circuitry **340** may receive a signal from a CPU of a host device (e.g., host device **114**), the multiplexer **326** and demultiplexer **328**, or other control circuitry that indicates which of the offload circuitries **102** is selected. The control circuitry **340** determines the grouping of the AFIFOs **324** and **330** based on the data bus width of the interface(s) **106** of the selected offload circuitry(s) **102** and provides control signals to the AFIFOs **324** and **330** to group the AFIFOs **324** and **330** accordingly. As different offload circuitries **102** are selected, the control circuitry **340** provides control signals that groups the AFIFOs **324** and **330** accordingly. In one or more examples, the control circuitry **340** is omitted, and the multiplexer **326** and demultiplexer **328** perform the grouping and connections between AFIFOs **324** and **330**. While the control circuitry **340** is illustrated as being within the processor circuitry **110**, in other examples, the control circuitry **340** may be disposed within the programmable logic **108**, or in a device external to the accelerator device **100** (e.g., the host device **114**).

In one or more examples, while AFIFOs **324** and **330** are described in the above (and in the following), other types of buffers may be used without deviating from the above description. For example, synchronous FIFOs, and last in

first out (LIFO) buffers, and data buffers that receive data in any order and output data in any order, among others (e.g., unordered buffer). In one or more example, in the above description AFIFOs **324** may be buffers and AFIFOs **330** may be buffers. The buffers may be any one or more of AFIFOs, FIFOs, synchronous FIFOs, LIFO buffers, and unordered buffers, among others.

Referring to FIG. 4, depicted is a schematic block diagram of a configuration of the AFIFOs **324** of the accelerator device **100**. In the example of FIG. 4, the offload circuitry **102b** is associated with an interface **106b** having interfaces **406a**, **406b**, **406c** and **406d**. The interfaces **406a**, **406b**, **406c**, and **406d** have 100, 256, 221 and 512 bit wide data buses, respectively. The multiplexer **326** couples the each of the interfaces **406a**, **406b**, **406c** and **406d** to an appropriate number of AFIFOs **324**. In one example, each of the AFIFOs **324** are 32 bits wide. Referring back to FIG. 3, likewise the demultiplexer **328** couples the interfaces **406a**, **406b**, **406c**, and **406d** to an appropriate number of AFIFOs **330**. In one example, the data width of the AFIFOs **330** is 32 bits wide.

For the offload circuitry **102b** and the interfaces **406a**, **406b**, **406c**, and **406d** having 100, 256, 221 and 512 bit wide data buses, thirty-five AFIFOs **324** are used. For example, four AFIFOs **324** are used for the interface **406a** to provide a data width of 100 bits, eight AFIFOs **324** are used for the interface **406b** to provide a data width of 256 bits, seven AFIFOs **324** are used for the interface **406c** to provide a data width of 221 bits, and sixteen AFIFOs **324** are used for the interface **406d** to provide a data width of 512 bits. Further, to support the interfaces **404a**, **404b** of interface **106a** of the offload circuitry **102b**, forty AFIFOs **324** are used. For example, twenty AFIFOs **324** are used for the interface **404a** to provide a data width of 640 bits, and twenty AFIFOs **324** are used for the interface **404b** to provide a data width of 640 bits. To support the interfaces **408a**, **408b**, **408c** of interface **106c** of the offload circuitry **102c**, twenty-five AFIFOs **324** are used. For example, five AFIFOs **324** are used for the interface **408a** to provide a data width of 160 bits, sixteen AFIFOs **324** are used for the interface **408b** to provide a data width of 512 bits, and four AFIFOs **324** are used for the interface **408c** to provide a data width of 128 bits. Accordingly, the largest number of AFIFOs **324** used to support the interfaces **106a**, **106b**, and **106c** is forty. Similarly, the AFIFOs **330** are grouped in a similar way to support the interconnections between the AFIFOs **324** and the programmed circuit blocks **104a**.

FIG. 7 illustrates a flowchart of a method **700** for coupling the AFIFOs **324**, **330** with the offload circuitries **102** and the programmed circuit blocks **104**. At **710** of the method **700**, the number of AFIFOs **324**, **330** are determined for grouping for a selected offload circuitry. For example, the control circuitry **340** determines the number and groupings of the AFIFOs **324**, **330** based on the selected offload circuitry. In one example, the number and groupings of AFIFOs **324**, **330** are determined to support the interfaces of the interfaces **106** based on the selected offload circuitry(s) **102**. The AFIFOs **324**, **330** are selected by control circuitry **340** connected to the AFIFOs **324**, **330** based on the interface **106** (and interfaces) selected by the multiplexer **326** and the demultiplexer **328**. In one example, an offload circuitry **102** is selected for use in processing data provided by a CPU of the host device **114**. In one example, an offload circuitry **102** is selected based on the type of data to be processed and/or task to be offloaded from the host device **114**. In one example, the host device **114** indicates which of the offload circuitries **102** to be selected. In one example with reference to FIG. 3 and FIG. 4, the offload circuitry **102b** is selected. The offload

circuitry 102b includes interface 106b and interfaces 406a, 406b, 406c, and 406d have a data bus width of 100 bits, 256 bits, 221 bits, and 512 bits, respectively. In one example, the control circuitry 340 determines the groupings of AFIFOs 324, 330 based on the data bus widths of the interfaces 406a, 406b, 406c, and 406d. For example, the control circuitry 340 determines that four AFIFOs 324 are used to support the interface 406a to provide a data width of 100 bits, eight AFIFOs 324 are used to support the interface 406b to provide a data width of 256 bits, seven AFIFOs 324 are used to support the interface 406c to provide a data width of 221 bits, and sixteen AFIFOs 324 are used to support the interface 406d to provide a data width of 512 bits. The control circuitry 340 groups the AFIFOs 324, 330 accordingly to support the interfaces 406a, 406b, 406c, and 406d. For example, the AFIFOs 324, 330 are grouped to accommodate the data bus widths for each of the interfaces of the interfaces 106 such that the combined data bus widths of the AFIFOs 324, 330 are at least as large as the data bus widths of the corresponding interfaces. In some examples, concatenating control and data transfers for each group of AFIFOs 324, 330 associated with respective ones of the interfaces 106. The number of AFIFOs 324, 330 is determined to accommodate (be at least as large as) all of the data bus widths of the interfaces of the selected offload circuitry 102.

At 720 of the method 700, first AFIFOs are coupled to output(s) of a multiplexer. For example, the control circuitry 340 couples a first one or more AFIFOs 324 to an output of the multiplexer 326. The one or more AFIFOs 324 corresponds to a first interface 106 of a selected offload circuitry 102. In an example where the first interface is interface 406a of FIG. 4, the control circuitry 340 couples four AFIFOs 324 to an output of the multiplexer 326. Further, for the interface 406b, eight AFIFOs 324 are coupled to an output of the multiplexer 326, for the interface 406c, seven AFIFOs 324 are coupled to an output of the multiplexer 326, and for the interface 406d, sixteen AFIFOs 324 are coupled to an output of the multiplexer 326.

In one or more examples, data is provided (e.g., pushed) to each AFIFO 324 of a first group of AFIFOs 324 based on a first clock cycle of a clock signal of the processor circuitry 110. Further, data is output (e.g., popped) from each AFIFO 324 of the first group of AFIFOs 324 based on a first clock cycle of a clock signal of the processor circuitry 110. Accordingly, data is loaded into each of the first group of the AFIFOs 324 during an overlapping period. In one example, each AFIFO 324 of a first group is provided with data based on each of the AFIFOs of the first group having data capacity (e.g., being empty or containing data and not being full). The control circuitry 340 determines whether or not that each of the AFIFOs having data capacity. In other examples, another circuitry of the accelerator device 100 determines whether or not that each of the AFIFOs have data capacity. In one or more examples, data is output (e.g., popped) from a group of the AFIFOs 324 to a group of the AFIFOs 330. Data is output from each AFIFO 324 of a first group based on each, or at a least one, of the AFIFOs 324 of the contain data. The control circuitry 340 determines whether or not that the AFIFOs 324 contain data, and data is output based on the determination that the AFIFOs 324 contain data. In other examples, another circuitry of the accelerator device 100 determines whether or not that the AFIFOs 324 contain data. Receiving data based on the AFIFOs 324 being determined to be empty and output data based on the AFIFOs 324 being determined to contain data mitigates skew that may occur within an individual AFIFO 324 when communicating data to the AFIFOs 330.

At 730 of the method 700, second AFIFOs are coupled to the first AFIFOs and to a demultiplexer. For example, the control circuitry 340 couples a first one or more AFIFOs 330 to outputs of the AFIFOs 324 and to an input of the demultiplexer 328. The one or more AFIFOs 330 corresponds to a first interface 106 of a selected offload circuitry 102. The grouped AFIFOs 330 are connected to a corresponding group of the AFIFOs 324. In an example where the first interface is interface 406a of FIG. 4, the control circuitry 340 couples four AFIFOs 330 to outputs of a group of four AFIFOs 324 and to an input of the demultiplexer 328. Further, for the interface 406b, eight AFIFOs 330 are coupled to outputs of a group of eight AFIFOs 324 and to an input of the demultiplexer 328, for the interface 406c, seven AFIFOs 324 are coupled to outputs of a group of seven AFIFOs 324 and to an input of the demultiplexer 328, and for the interface 406d, sixteen AFIFOs 330 are coupled to outputs of a group of sixteen AFIFOs 324 and to an input of the demultiplexer 328. Accordingly, by performing the method 700 of FIG. 7, an interface is provided from a selected offload circuitry 102 through the multiplexer 326, the AFIFOs 324, 330, and the demultiplexer 328 to the programmed circuit blocks 104, such that the data bus width of the interface is supported by the number of grouped AFIFOs 324 and 330.

In one or more examples, data is provided (e.g., pushed) to each AFIFO 330 of a first group of AFIFOs 330 based on the same clock cycle of a clock signal of the programmable logic 108. Further, data is output (e.g., popped) from each AFIFO 330 of the first group of AFIFOs 330 based on the same clock cycle of a clock signal of the processor circuitry 110. In one example, each AFIFO 330 of a first group is provided with data based on each of the AFIFOs 330 of the first group have data capacity. The control circuitry 340 determines whether or not that each of the AFIFOs 330 have data capacity. In other examples, another circuitry of the accelerator device 100 determines whether or not that each of the AFIFOs 330 have data capacity. Further, data is output from each AFIFO 324 of a group based on each, or at least one, of the AFIFOs of the group containing data. The control circuitry 340 determines whether or not that the AFIFOs 330 containing data. In other examples, another circuitry of the accelerator device 100 determines whether or not that the AFIFOs 330 contain data. Further, the data is output (e.g., popped) from the group or groups of the AFIFOs 330 to the demultiplexer 328.

In one or more examples, the AFIFOs 330 are synchronized with the AFIFOs 324 via credit signals. For example, internal counters of the credits are maintained to indicate the AFIFO depth (e.g., outstanding requests holding capacity). The counters are decremented every time a valid signal and data is asserted. The valid signal is deasserted if there are not enough credits in the corresponding credit counter. In one example, a data transmitter (e.g., an offload circuitry 102) increments the credit counter every time a credit increment is received from the data receiver (e.g., a programmed circuit block 104). As is described in further detail in the following, this allows the valid signal to be decoupled from the ready signal and allows for adding repeater flip-flops (source synchronous) on the valid/data and ready signals independently.

Referring to FIG. 6, depicted is a schematic logic block diagram of credit-based data transfer circuits, according to one or more specific example embodiments of this disclosure. A fixed size AFIFO comprising a plurality of AFIFOs 324, 330 are used and instead of providing a Ready within a data transfer cycle use credits in combination with valid

11

and ready control. The AFIFOs 324 provide credit returns to the data transmitters (e.g., offload circuitries 102 of FIG. 1) of multiplexer 326. The AFIFOs 330 provide credit returns to the data transmitters of AFIFOs 324. The demultiplexer 328 provide credit returns to the data transmitters of AFIFOs 330.

The data transmitters maintain internal counters of the credits (FIFO depth) and decrement the counter every time a data transmitter asserts valid and data signals. The data transmitter de-asserts the valid signal if there are not enough credits in its credit counter. The data transmitter credit counter increments every time a credit increment is received from the data receiver (e.g., the programmed circuit block 104). This allows the valid signal to be decoupled from the ready signal and provides for adding repeater flip-flops 640-646 (source synchronous) on the valid/data and Ready signals independently. The repeater flip-flops 640-646 are included to meet timing constraints between the AFIFOs 324 and the AFIFOs 330. For example, the repeater flip-flops 640-646 are used to control the timing of data that is transmitted between the AFIFOs 324 and the AFIFOs 330. It is contemplated and within the scope of this disclosure that other forms and methods of data transfer control may be used with equal effectiveness, and one having ordinary skill in the art of digital logic and the benefit of this disclosure could design such other data transfer control circuits.

While the foregoing is directed to embodiments of the present invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. An interconnection circuitry comprising:
 - a multiplexer configured to couple to first offload circuitry and receive data therefrom;
 - a first plurality of buffers having inputs coupled to outputs of the multiplexer, wherein a first buffer group of the first plurality of buffers is connected to the first offload circuitry via the multiplexer, and a number of buffers of the first plurality of buffers within the first buffer group corresponds to a data width associated with the first offload circuitry and is less than a total number of buffers within the first plurality of buffers;
 - a second plurality of buffers comprising inputs coupled to outputs of the first plurality of buffers; and
 - a demultiplexer comprising inputs coupled to outputs of the second plurality of buffers and outputs coupled to inputs of programmable logic.
2. The interconnection circuitry of claim 1, wherein each of the first plurality of buffers comprises a first data bus width, and wherein each of the second plurality of buffers has a second data bus width.
3. The interconnection circuitry of claim 2, wherein the first data bus width is equal to the second data bus width.
4. The interconnection circuitry of claim 1, wherein a combined data bus width of the first buffer group is at least equal to a data bus width of the first offload circuitry.
5. The interconnection circuitry of claim 4, wherein a combined data bus width of the second plurality of buffers is at least equal to the data bus width of the first offload circuitry.
6. The interconnection circuitry of claim 1, wherein the multiplexer is further configured to couple to second offload circuitry and receive data therefrom, wherein a second buffer group of the first plurality of buffers is connected to the second offload circuitry via the multiplexer, and wherein the second plurality of buffers has inputs coupled to outputs of

12

the second buffer group, and wherein the inputs of the demultiplexer are coupled to outputs of the second plurality of buffers.

7. The interconnection circuitry of claim 6, wherein a number of buffers of the second buffer group differs from the number of buffers of the first buffer group.

8. The interconnection circuitry of claim 1, wherein data is provided to each buffer of the first buffer group during a clock cycle of a first clock signal and based on the first buffer group having data capacity, and data is output from each buffer of the first buffer group during a clock cycle of a second clock signal and based on the first buffer group containing data, wherein the first clock signal differs from the second clock signal.

9. An accelerator device comprising:

first offload circuitry; and

interconnection circuitry coupled to the first offload circuitry, the interconnection circuitry comprising:

a multiplexer coupled to the first offload circuitry;

a first plurality of buffers comprising inputs coupled to outputs of the multiplexer, wherein a first buffer group of the first plurality of buffers is connected to the first offload circuitry via the multiplexer, and a number of buffers of the first plurality of buffers within the first buffer group corresponds to a data width associated with the first offload circuitry and is less than a total number of buffers within the first plurality of buffers;

a second plurality of buffers comprising inputs coupled to outputs of the first plurality of buffers; and

a demultiplexer comprising inputs coupled to outputs of the second plurality of buffers and outputs configured to couple to inputs of programmable logic.

10. The accelerator device of claim 9, wherein each of the first plurality of buffers comprise a first data bus width, and wherein each of the second plurality of buffers comprise a second data bus width, wherein the first data bus width is equal to the second data bus width.

11. The accelerator device of claim 9, wherein a combined data bus width of the first buffer group is at least equal to a data bus width of the first offload circuitry.

12. The accelerator device of claim 11, wherein a combined data bus width of the second plurality of buffers is at least equal to the data bus width of the first offload circuitry.

13. The accelerator device of claim 9 further comprising second offload circuitry, wherein the multiplexer is further configured to couple to the second offload circuitry, wherein a second buffer group of the first plurality of buffers is connected to the second offload circuitry via the multiplexer, and wherein the second plurality of buffers has inputs coupled to outputs of the second buffer group, and wherein the inputs of the demultiplexer are coupled to outputs of the second plurality of buffers.

14. The accelerator device of claim 13, wherein a number of buffers of the second buffer group differs from the number of buffers of the first buffer group.

15. The accelerator device of claim 9, wherein data is provided to each buffer of the first buffer group during a clock cycle of a first clock signal and based on the first buffer group having data capacity, and data is output from each buffer of the first buffer group during a clock cycle of a second clock signal and based on the first buffer group containing data, wherein the first clock signal differs from the second clock signal.

16. A method comprising:

determining, based on a data bus width of a first offload circuitry, a first number of a first plurality of buffers

13

within a first buffer group and a second number of a second plurality of buffers within a second buffer group, wherein the first number corresponds to a data width associated with the first offload circuitry and is less than a total number of buffers within the first plurality of buffers;

coupling inputs of the first plurality of buffers to outputs of a multiplexer, wherein inputs of the multiplexer are coupled to outputs of the first offload circuitry; and

coupling inputs of the second plurality of buffers to outputs of the first plurality of buffers and coupling outputs of the second plurality of buffers to inputs of a demultiplexer, wherein outputs of the demultiplexer are coupled to inputs of programmable logic.

17. The method of claim **16**, wherein each of the first plurality of buffers comprises a first data bus width, and each of the second plurality of buffers comprises a second data bus width, wherein the first data bus width is equal to the second data bus width.

18. The method of claim **16**, wherein a combined data bus width of the first buffer group is at least equal to the data bus width of the first offload circuitry, and wherein a combined data bus width of the second buffer group is at least equal to the data bus width of the first offload circuitry.

14

19. The method of claim **16** further comprising: determining, based on a data bus width of a second offload circuitry, a number of a third buffer group of the first plurality of buffers and a number of a fourth buffer group of the second plurality of buffers, wherein a number of the third buffer group differs from a number of the fourth buffer group;

coupling inputs of the third buffer group to outputs of the multiplexer, wherein inputs of the multiplexer are coupled to outputs of the second offload circuitry; and

coupling inputs of the fourth buffer group to outputs of the third buffer group and coupling outputs of the fourth buffer group to the inputs of the demultiplexer, the number of the third buffer group differs from the first number of the first buffer group.

20. The method of claim **16** further comprising: providing data to each of the first plurality of buffers during a clock cycle of a first clock signal and based on the first plurality of buffers having data capacity; and outputting data from each of the first plurality of buffers during a clock cycle of a second clock signal and based on the first plurality of buffers containing data, wherein the first clock signal differs from the second clock signal.

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